

US Debt Tracker

How fast US public debt grew between 1993 and 2023, when it grows fastest within a year, and where the trend points.

Tools Excel. Pivot tables, charts, trend forecasting

Data US Treasury "Debt to the Penny". 7,495 daily records, April 1993 to February 2023

The questions

A debt analysis agency needed three questions answered for a client brief:

- What was the yearly percentage increase in debt compared to the previous year?
- Which months historically see the highest and lowest increases in total debt?
- What is the projected growth of publicly held debt over the next few years?

The data

Field	Meaning
Debt Held by the Public	Debt held by individuals, corporations, foreign governments and other entities outside the US government
Intragovernmental Holdings	Debt held by other US government agencies
Total Public Debt Outstanding	The sum of the two

The raw file arrived in wide format, 7,496 columns across with one per date, which is unusable for analysis. Restructuring it into a long table of date and value rows was the first step. The workbook keeps Raw Data and Cleaned Data as separate sheets so the transformation is visible.

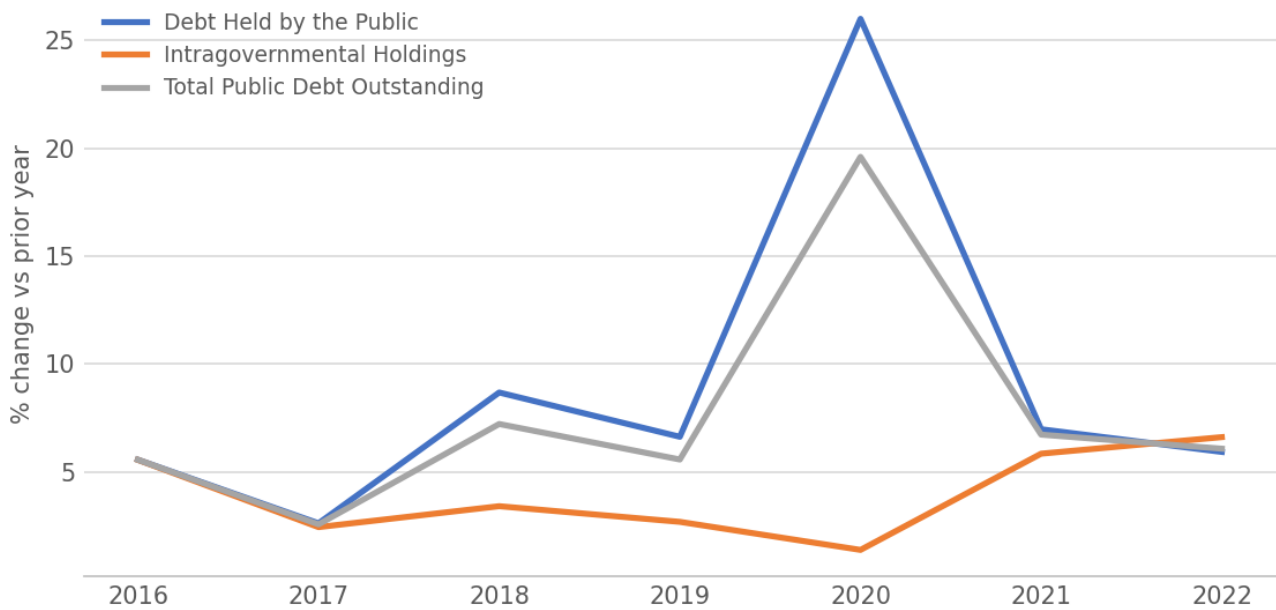
How I approached it

- Transposed the raw wide export into a clean date-indexed table
- Built year-over-year percentage change across all three debt measures
- Used a pivot table to average total debt by calendar month across 30 years, isolating seasonal pattern from long-term growth
- Applied trend forecasting to project publicly held debt through 2027
- Charted each question and pulled the conclusions onto a single output sheet

Yearly growth

Between 2016 and 2019 total public debt rose by an average of 5% a year. 2020 breaks the pattern sharply. Publicly held debt jumped 26% in a single year, consistent with the scale of US pandemic response spending. Intragovernmental holdings barely moved, so the entire spike sits on the public side of the ledger.

Yearly Debt Percentage Increase

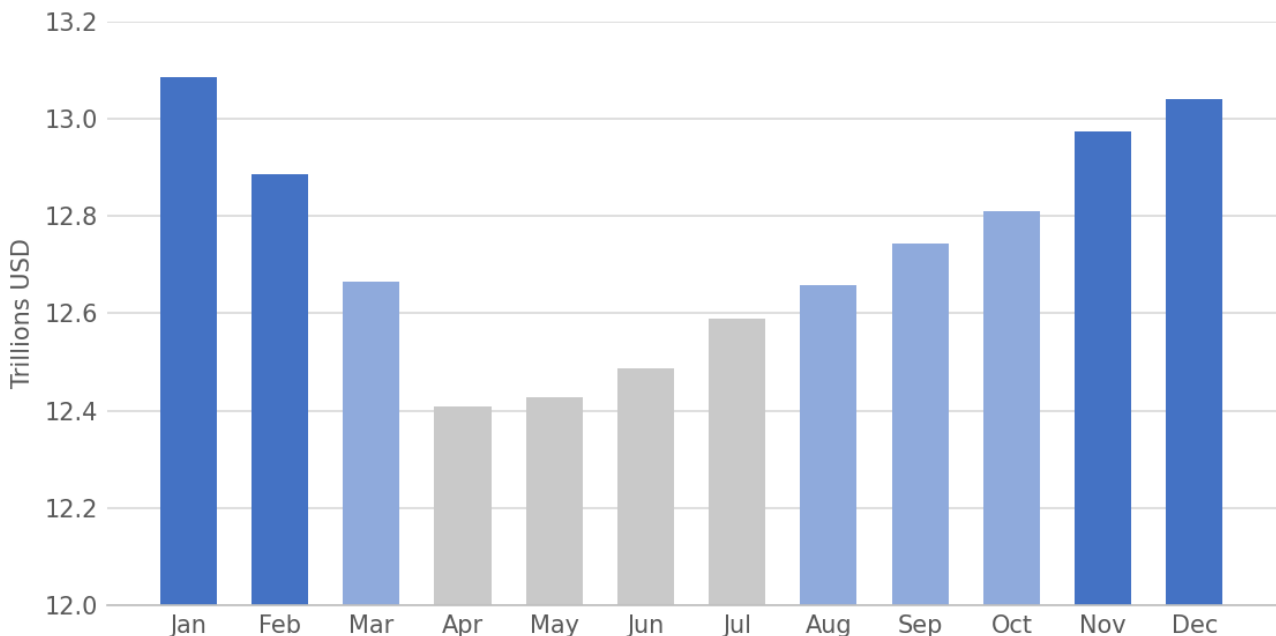


Year-over-year percentage change by debt measure, 2016 to 2022.

Seasonality

Averaged across three decades, the largest debt levels land in January, February, November and December. The lowest fall in April, May, June and July, which lines up with tax receipts arriving in the spring.

Average Total Debt by Calendar Month



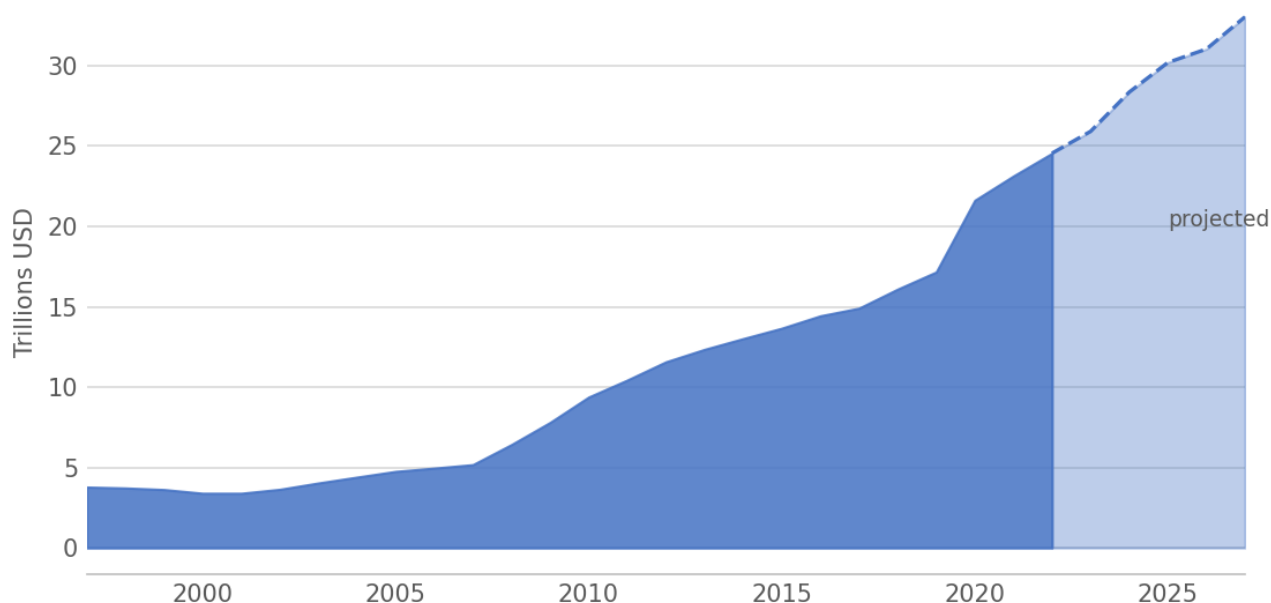
Average total public debt by calendar month, 1993 to 2023. Blue marks the four highest months, grey the four lowest.

Long-term trajectory

Publicly held debt took a decade to add its first trillion, from roughly \$3.8T in 1997 to \$5.2T in 2007. It then went from \$6.4T in 2008 to \$17.2T in 2019, and reached \$24.6T by early 2023. On the current trend it

reaches about \$33T by 2027.

Publicly Held Debt, Actual and Projected



Publicly held debt, actual to 2022 and trend-projected to 2027.

Interpretation and limitations

The seasonal finding is the practical one. Anyone forecasting debt levels or timing debt-sensitive decisions should not read a January or February increase as a change in trend, because that is the normal shape of the year. April through July is where a genuine slowdown would show up.

The projection is a straight trend extrapolation and should be read as such. It assumes no policy change, no rate shock and no repeat of a 2020-scale event, and 2020 itself shows how quickly that assumption breaks. It is a baseline to measure against, not a forecast to rely on.

What's in the workbook

Sheet	Contents
Scenario	The brief and data dictionary
Raw Data	The original file, untouched
Cleaned Data	Restructured long-format table
Yearly Percentage Change	Year-over-year calculation and chart
Monthly Historicals	Monthly pivot and seasonal chart
Projected Growth	Annual maximums, forecast and chart
Final Outputs	All three charts and written conclusions

Portfolio project built on public US Treasury data. The agency client in the brief is a scenario used to frame the analysis, not a real engagement.